Literature review

Efficient modelling of time-domain dynamic positioning simulations

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1 Introduction

In this section the background and scope of the literature study will be presented. First some background information on dynamic positioning industry practices will be given followed by a motivation for the thesis that will follow. This will culminate in the research question and subsequent goals of the thesis.

1.1. Background

Since the introduction of the first ship equipped with dynamic positioning in the 1960's, the use of dynamic positioning has become much more widespread in the offshore industry.

1.2. Previous work

- Increased use of dynamic DP simulations - Deterministic DP simulations, no margin for varying input parameters - Time consuming simulations

- Probabilistic approach to capability - calculated capability for different Hs/Tp/Vw combinations, then determined most probable capability based on chance of each Hs/Tp/Vw combination occuring Mauro and Nabergoj, 2022 - Hydrodynamic load estimation using PINNs, used as added mass etc isn't perfect. Estimateed based on measured motions. Ding et al., 2025 - Simulation of pipe coupled with DP Ai et al., 2018, Armaoğlu and Monti, 2014, Li-ping et al., 2014

- Developed a fast method for evaluating the DP capability in time domain for early design. Not sure how it is fast though, this isn't discussed Luebcke et al., 2015

- Serraris and Cozijn, n.d. performed DP capability analyses with aNySIM from MARIN. performed 8700 simulations of each 10 minutes approximately to determine capability under various conditions. Only had 9 headings

- Following studies quantified size of their modelling errors. All used deterministic models: - Børhaug, 2012, comparison to model tests. 10% error in power demand - Martelli et al., 2022 performed model tests, compared static capability to dynamic. Non-linear factor differences, changed for each condition - **donnarummaNUMERICALMODELSSHIP2015** compared high fidelity model to low fidelity, two different power trains. Much larger power peaks, peaks 25% to 50% power, wheras simplified had constant 20% consumption

These errors tended to be quite large for use in decision making.

Hop, 2024 and De Vries, 2020 modelled pipelay dynamics.

- System identification of non-linear systems is a large and relatively known field - Create model of mainly unknown system to predict behaviour.

- Tsiakalos and Tsiakalos, 2026 propose a pipeline to predict power grid consumption while subject to exogenous variables such as weather conditions. They use ML to make the predictions as well as

quantify the uncertainty of the prediction. - Used UMAP to reduce input dimensionality, then performed clustering to identify patterns for different households - Train a forecaster for each cluster to allow it to specialise more. - uncertainty was based on the forecasters, as these output quantiles or distributions. The more certain the forecaster was, the smaller the uncertainty. - Main goal was to used the clustering to improve accuracy for longer forecasts compared to a global forecaster - Relevance to DP -> Dynamic system subject to exogenous variables. Quantifies uncertainty of prediction

- Ebadollahi and Rahman, 2025 propose a method using NARX (Nonlinear Auto-Regressive with eXogenous input) to perform uncertainty quantification on non-linear dynamic systems. As an example they used a pickup truck travelling over bumps. - Used PDD (Polynomial dimensional decomposition) to propegate uncertainty

- Salinas et al., 2020 discusses a recurrent neural network architecture to create probabilistic forecasts with uncertainty. - Introduces DeepAR, a probabilistic forecasting method using autoregressive recurrent neural networks trained on large sets of related time series for improved accuracy and scalability. - Handles scale variability through rescaling and weighted sampling, and supports flexible likelihood functions (Gaussian for real-valued data, Negative Binomial for count data). - Generates full predictive distributions via Monte Carlo sampling, outperforming traditional methods in both point and probabilistic forecasts across multiple datasets.

- Yogeesh et al., 2025 predict vessel motions using fuzzy-set logic - Uses deterministic calculations over an interval to predict effect of uncertain inputs. -

So far not covered: How DP behaviour is influenced by varying parameters, how sensitive is it to uncertainty? Probabilistic dynamic DP perediction, quantified uncertainty (how good is the prediction likely to be?)

1.3. Research question

How can a pipelay ship's short term dynamic positioning behaviour efficiently be forecast while considering environmental and system uncertainty?

To answer this question, the following sub-questions will be investigated:

1. What simulation framework gives the best trade-off between accuracy and computational efficiency?
2. How can uncertainty in model parameters and external inputs be quantified and propagated through the forecasting model
3. How does the forecasting model perform compared to traditional DP simulation methods in terms of speed and accuracy?

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